> Genes are selected, not as 'good' in isolation, but as good at working against the background of other genes in the gene pool. A good gene must be compatible with, and complementary to, the other genes with whom it has to share a long succession of bodies. (p. 110)

I have written a fuller reply to criticisms of genetic atomism in *The Extended Phenotype*, especially on pp. 116–17 and 239–47.

***p. 36* The definition I want to use comes from G. C. Williams.**

Williams's exact words, in *Adaptation and Natural Selection*, are:

> I use the term gene to mean 'that which segregates and recombines with appreciable frequency.' … A gene could be defined as any hereditary information for which there is a favorable or unfavorable selection bias equal to several or many times its rate of endogenous change.

Williams's book has now become widely, and rightly, regarded as a classic, respected by 'sociobiologists' and critics of sociobiology alike. I think it is clear that Williams never thought of himself as advocating anything new or revolutionary in his 'genic selectionism', and no more did I in 1976. We both thought that we were simply reasserting a fundamental principle of Fisher, Haldane, and Wright, the founding fathers of 'neo-Darwinism' in the 1930s. Nevertheless, perhaps because of our uncompromising language, some people, including Sewall Wright himself, apparently take exception to our view that 'the gene is the unit of selection'. Their basic reason is that natural selection sees organisms, not the genes inside them. My reply to views such as Wright's is in *The Extended Phenotype*, especially pp. 238–47. Williams's most recent thoughts on the question of the gene as the unit of selection, in his 'Defense of Reductionism in Evolutionary Biology', are as penetrating as ever. Some philosophers, for example, D. L. Hull, K. Sterelny and P. Kitcher, and M. Hampe and S. R. Morgan, have also recently made useful contributions to clarifying the issue of the 'units of selection'. Unfortunately there are other philosophers who have confused it.

***p. 43* … the individual is too large and too temporary a genetic unit …**

Following Williams, I made much of the fragmenting effects of meiosis in my argument that the individual organism cannot play the role of replicator in natural selection. I now see that this was only half the story. The other half is spelled out in *The Extended Phenotype* (pp. 97–9) and in my paper 'Replicators and Vehicles'. If the fragmenting effects of meiosis were the whole story, an asexually reproducing organism like a female stick-insect would be a true replicator, a sort of giant gene. But if a stick insect is changed—say it loses a leg—the change is not passed on to future generations. Genes alone pass down the generations, whether reproduction is sexual or asexual. Genes, therefore, really are replicators. In the case of an asexual stick-insect, the entire genome (the set of all its genes) is a replicator. But the stick-insect itself is not. A stick-insect body is not moulded as a replica of the body of the previous generation. The body in any one generation grows afresh from an egg, under the direction of its genome, which *is* a replica of the genome of the previous generation.

All printed copies of this book will be the same as one another. They will be replicas but not replicators. They will be replicas not because they have copied one another, but because all have copied the same printing plates. They do not form a lineage of copies, with some books being ancestral to others. A lineage of copies would exist if we xeroxed a page of a book, then xeroxed the xerox, then xeroxed the xerox of the xerox, and so on. In this lineage of pages, there really